

Annie Prasanna Manoharan

+1(479)301-1366 | annieprm0501@gmail.com | <https://annie0159.github.io/Portfolio/> | <https://github.com/Annie0159>

EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Bachelor of Engineering in Electronics and Communication, India

Aug 2019 – May 2023

Sri Ramakrishna Engineering College

TECHNICAL SKILLS

Embedded Systems: Embedded C, C, C++, C#, Python, Microcontrollers, Microprocessors, Firmware Development, Real-Time Systems, FreeRTOS, Hardware/Software Co-Design, Interrupts, Timers, ADC, GPIO, PWM, Sensor/Actuator Control

Digital Hardware: FPGA, SoC, Xilinx Arty, MicroBlaze, VHDL, Verilog, Assembly, RTL/Digital Logic, AXI-Based IP Cores, Hardware Description Languages, Quad-SPI, SREC Bootloader, Vivado, Vitis

Communication Interfaces: UART, SPI, I2C, MQTT, AXI, Serial Communication, Sensor Interfaces, Inter-Module Communication

Verification & Validation: Hardware Testing, Test Automation, TestStand, LabVIEW, MAVIS, Test Case Development, Regression Testing, Debugging, Test Coverage, Medical Device Testing, Quality Assurance

Sensors, Controls & Instrumentation: Pressure Sensors, Infrared Sensors, Ultrasonic Sensors, Temperature Sensors, Data Acquisition, PID Control, Feedback Control, Servo Control, Motor Control, Real-Time Control, Autonomous Systems

Development Tools: Git, Bitbucket, Jira, Agile, Scrum, SDLC, Object-Oriented Programming

PROFESSIONAL EXPERIENCE

Project Engineer, Soliton Technologies, India

June 2023 – June 2024

- Developed a C#/NET WPF desktop application for Dräger integrating C#, Python, and LabVIEW-based test automation, improving test workflow efficiency, user onboarding, and automated test coverage for medical-device engineering
- Developed customized automated testing solutions using MAVIS, an object-oriented test automation framework built on NI LabVIEW and Python, supporting verification and validation (V&V) activities for medical-device applications
- Served as the sole developer on a Medtronic project, automating approximately 94% of manual test cases while supporting quality and compliance requirements in a medical-device development environment
- Integrated software components and test automation workflows across C#, Python, LabVIEW, and TestStand, demonstrating hardware/software integration and automated validation capabilities

Engineering Intern, Soliton Technologies, India

July 2022 – May 2023

- Received intensive hands-on training in LabVIEW, Object-Oriented Programming, TestStand, and automated testing, developing applications for verification and validation workflows
- Worked with MAVIS, a LabVIEW/Python-based test automation framework, to streamline automated testing processes for medical-device engineering projects

Embedded Systems Intern, Crystal Clear Technology and Innovation, India

July 2020

- Developed embedded applications using the PIC16F887 microcontroller, working with timers, ADC, LCD interfacing, UART, SPI, and I2C
- Designed and simulated microcontroller-based circuits using MPLAB IDE and Proteus, gaining hands-on experience in embedded firmware development and circuit simulation

PROJECTS

FPGA-Based Autonomous Vehicle Xilinx

Arty, MicroBlaze, Vivado, Vitis

- Designed and implemented an autonomous line-following vehicle using a Xilinx Arty FPGA and custom MicroBlaze-based SoC
- Integrated infrared line sensors, ultrasonic sensors, and a motor driver through AXI-based IP cores in Vivado
- Developed embedded control software in Vitis for real-time line tracking, obstacle detection, and motor control
- Implemented a custom boot process using Quad-SPI flash memory and an SREC bootloader, enabling autonomous application startup after power-up
- Demonstrated FPGA system design, hardware/software co-design, embedded programming, peripheral integration, and autonomous robotics

Elevator Control System

PIC16F887, Embedded C, Proteus, MPLAB IDE

- Designed and implemented a multi-floor elevator control system using the PIC16F887 microcontroller and Embedded C, implementing a finite state machine (FSM) for floor selection, position detection, movement, and real-time control.
- Implemented bidirectional motor control using relay circuits and digital sensor inputs, with embedded logic for accurate floor-position detection and elevator movement.
- Integrated LCD and 7-segment displays for real-time floor/status indication and simulated and debugged the embedded system using Proteus and MPLAB IDE.